마이크로플루이딕 기반 세포 내 전달 플랫폼: 기술 발전, 응용 및 상용화

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Microfluidic Intracellular Delivery Platforms: Advancements, Applications, and Commercialization

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The efficient internalization of exogenous biomolecules into living cells enables cell engineering and supports advances in both basic biological research and clinical translation. Conventional delivery technologies including viral vectors, cationic lipids and electroporation often involve trade-offs between delivery efficiency and the preservation of cell viability, phenotype and function.

To address these limitations, we have developed a series of next-generation microfluidic platforms for intracellular delivery. By exploiting fluid cell interactions within precisely confined microchannels, these systems induce transient and reversible discontinuities in the cell membrane and nuclear envelope, enabling convective transport of biomolecules directly into the cell interior. Using this strategy, we have achieved highly efficient and functional delivery across a broad range of cell types, including human primary stem cells and immune cells.

In this talk, I will highlight recent scientific advances in microfluidic intracellular delivery and demonstrate their applications in genome editing and cancer immunotherapy. I will then discuss our ongoing efforts to commercialize these technologies through the lab spinout company, MxT Biotech, and share insights from translating microfluidic innovations from the laboratory to real world biomedical applications.

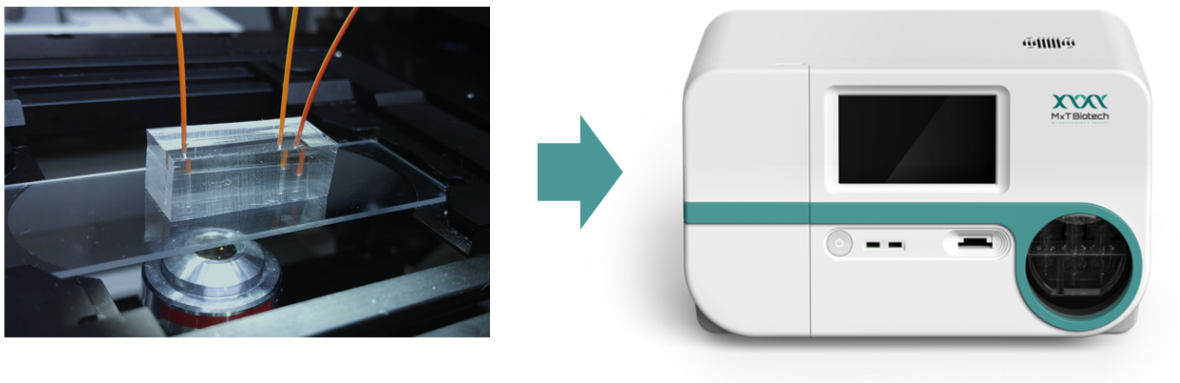

Fig. 1. (Left) A PDMS-based microfluidic intracellular delivery chip. (Right) The commercialized Hydroporator® device.